

Partially Ordered Relations :-

Relations and their properties :

Binary Relation :

Let A, B be any sets. A binary relation R from A to B , written $R : A \times B$ is a subset of the set $A \times B$.

Complementary Relation :

Let R be the binary relation from A to B . Then the complement of R can be defined by

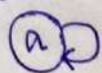
$$\bar{R} = \{(a, b) / (a, b) \notin R\} = (A \times B) - R$$

Inverse Relation :

Let R be the binary relation from A to B . Then the inverse of R can be defined by

$$R^{-1} = \{(b, a) / (a, b) \in R\}$$

Reflexive: A relation R on a set A is called reflexive if $(a, a) \in R$ for every element $a \in A$.



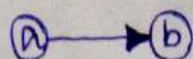
Every vertex has a self-loop.

Symmetric: A relation R on a set A is called symmetric if $(b, a) \in R$ whenever $(a, b) \in R$, for all $a, b \in A$.



If there is an edge from one vertex to another, there is an edge in the opposite direction.

Antisymmetric: A relation R on a set A such that for all $a, b \in A$, if $(a, b) \in R$ and $(b, a) \in R$, then $a = b$ is called antisymmetric.



Transitive: A relation R on a set A is called transitive whenever $(a, b) \in R$ and $(b, c) \in R$, then $(a, c) \in R$.

NOTE: A relation is partially ordered if it satisfies reflexive, anti-symmetry, & transitive properties.

The set A w.r.t partial ordered relation R is called POSET.

Example:

1. Which of the following relations on $\{0, 1, 2, 3\}$ are equivalence relations? Find properties of an equivalence relation that others lack.

(a) $R_1 = \{(0, 0), (1, 1), (2, 2), (3, 3)\}$

(b) $R_2 = \{(0, 0), (0, 2), (2, 0), (2, 2), (2, 3), (3, 2), (3, 3)\}$

(c) $R_3 = \{(0, 0), (1, 1), (1, 2), (2, 1), (2, 2), (3, 3)\}$

(d) $R_4 = \{(0, 0), (1, 1), (1, 3), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)\}$

(e) $R_5 = \{(0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (1, 2), (2, 0), (2, 2), (3, 3)\}$

Soln:

(a) R_1 is reflexive, symmetric, transitive.

$\therefore R_1$ is an equivalence relation.

(b) R_2 is reflexive, symmetric but not transitive, since $(3, 2)$

& $(2, 0) \in R_2$, but $(3, 0) \notin R_2$.

$\therefore R_2$ is not an equivalence relation.

(c) R_3 is reflexive, symmetric & transitive.

$\therefore R_3$ is an equivalence relation.

(d) R_4 is reflexive, symmetric but not transitive, since

$(1, 3)$ and $(3, 2) \in R_4$, but $(1, 2) \notin R_4$.

$\therefore R_4$ is not an equivalence relation.

(e) R_5 is reflexive, ^{but not} symmetric since $(1, 2) \in R_5$ but $(2, 1)$

$\notin R_5$. Also, R_5 is not transitive, since $(2, 0)$ & $(0, 1)$

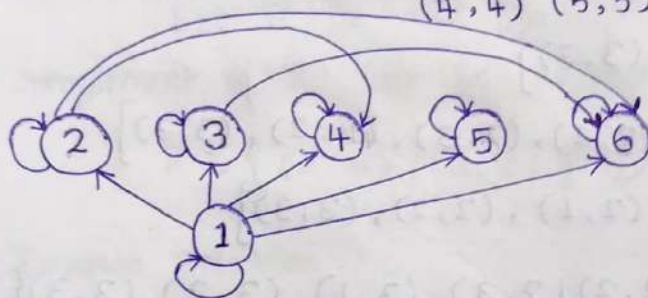
$\in R_5$, but $(2, 1) \notin R_5$.

$\therefore R_5$ is not an equivalence relation.

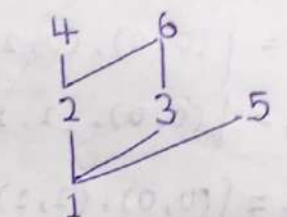
1. Draw the digraph representing the partial ordering on $\{(a, b) \mid a \text{ divides } b\}$ on the set $\{1, 2, 3, \dots, 6\}$ reduce the same to Hasse diagram.

Soln. : Let $A = \{1, 2, 3, 4, 5, 6\}$

$$R = \{(1, 1) (1, 2) (1, 3) (1, 4) (1, 5) (1, 6) (2, 2) (2, 4) (2, 6) (3, 3) (3, 6) (4, 4) (5, 5) (6, 6)\}$$



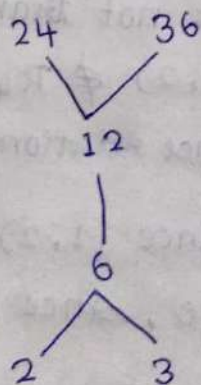
DI-GRAPH



HASSE DIAGRAM

2. Construct the Hasse diagram for the following set $A = \{2, 3, 6, 12, 24, 36\}$ which satisfies the relation $R = \{(a, b) \mid a \leq b\}$ where $\leq$ indicates a divides b .

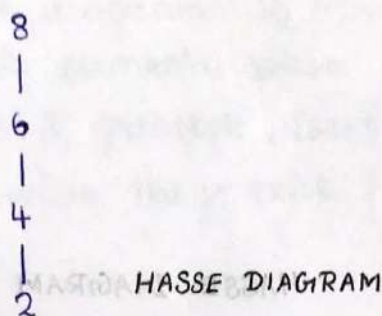
Soln. : $R = \{(2, 6) (2, 12) (2, 24) (2, 36) (2, 2) (6, 6) (6, 12) (6, 24) (6, 36) (12, 12) (12, 24) (12, 36) (24, 24) (36, 36)\}$



HASSE DIAGRAM

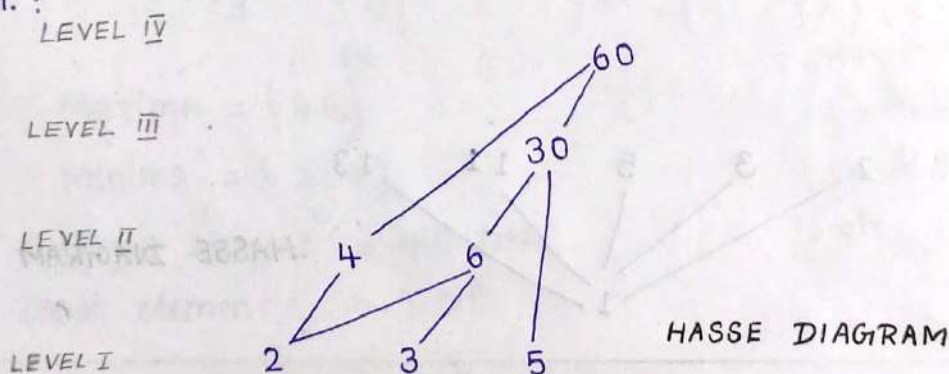
3. Draw the Hasse diagram for $(P, \leq)$ $P = \{2, 4, 6, 8\}$ where $\leq$ define "less than or equal to".

Soln.:



4. Draw the Hasse diagram for $(A, |)$ where $A = \{2, 3, 4, 5, 6, 30, 60\}$

Soln.:



Write if asked in the an

Maximal element = $\{60\}$

Minimal elements = $\{2, 3, 5\}$

Least element = $\emptyset$ not exist

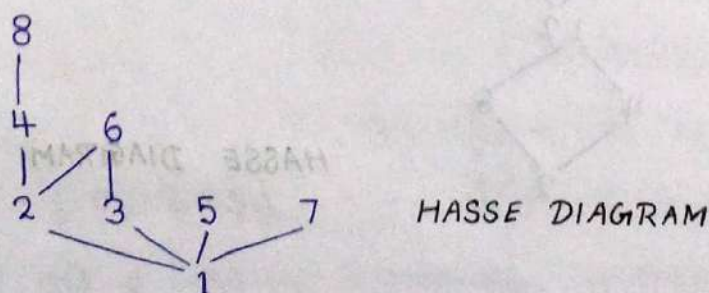
Greatest element = 60

Exercise:

1. Draw a Hasse diagram for $(A, |)$ (divisibility relation), where

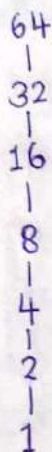
i) $A = \{1, 2, 3, 4, 5, 6, 7, 8\}$

Soln.:



ii) $A = \{1, 2, 4, 8, 16, 32, 64\}$

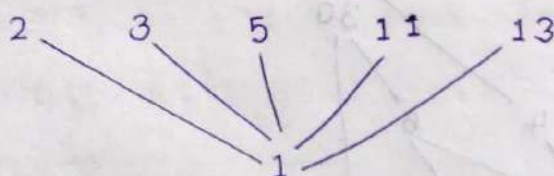
Soln.:



HASSE DIAGRAM

iii) $A = \{1, 2, 3, 5, 11, 13\}$

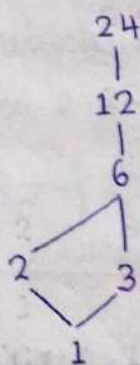
Soln.:



HASSE DIAGRAM

iv) $A = \{1, 2, 3, 6, 12, 24\}$

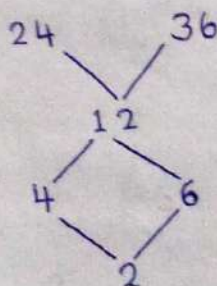
Soln.:



HASSE DIAGRAM

v) $A = \{2, 4, 6, 12, 24, 36\}$

Soln.:



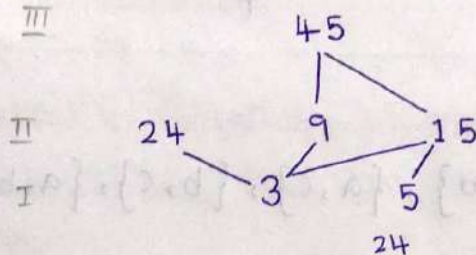
HASSE DIAGRAM

2. Consider the poset $(\{3, 5, 9, 15, 24, 45\}, \mid)$, that is, the divisibility relation.

- i) Draw its Hasse diagram ii) Find its maxima, minima, greatest and least elements when they exist. iii) Find maxima, minima & greatest, least elements of the set $M = \{3, 9, 15\}$, when they exist.

Soln. :

i)



HASSE DIAGRAM

ii)

$$\text{Maxima} = \{45\}$$

$$\text{Minima} = \{3, 5\}$$

$$\text{Greatest element} = 45 \text{ DNE}$$

$$\text{Least element} = \text{DNE}$$

Maximal elt - no further growth from the particular elt

iii)

$$M = \{3, 9, 15\}$$

$$\text{Maxima} = \{9, 15\}$$

$$\text{Minima} = \{3\}$$

$$\text{Greatest element} = \text{DNE}$$

$$\text{Least element} = 3$$

27/12/25

3. Draw the Hasse diagram representing the partial ordering $\{(A, B) \mid (A \subseteq B)\}$ on the power set $\mathcal{P}(S)$, where $S = \{a, b, c\}$. Find the maximal, minimal, greatest & least elements of the poset.

- i) $S = \{a\}$ ii) $S = \{a, b\}$ iii) $S = \{a, b, c\}$

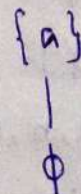
Soln. :

$$i) S = \{a\}$$

$$\mathcal{P}(S) = \{\emptyset, \{a\}\}$$

$$\text{Maximal elt} = \{a\}$$

$$\text{Minimal elt} = \{\emptyset\}$$



$$\text{Greatest} = \{a\}$$

$$\text{Least} = \phi$$

$$\text{ii) } S = \{a, b\}$$

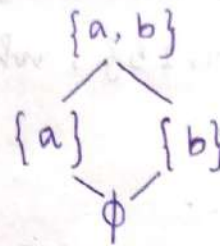
$$P(S) = \{\phi, \{a\}, \{b\}, \{a, b\}\}$$

$$\text{Maximal} = \{\{a, b\}\}$$

$$\text{Minimal} = \phi$$

$$\text{Greatest} = \{a, b\}$$

$$\text{Least} = \phi$$



$$\text{iii) } S = \{a, b, c\}$$

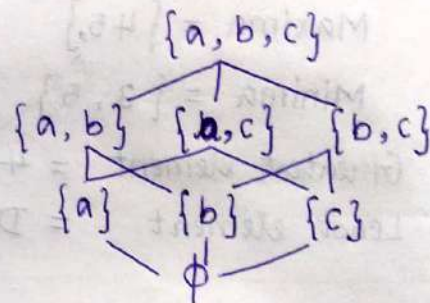
$$P(S) = \{\phi, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}$$

$$\text{Maximal} = \{\{a, b, c\}\}$$

$$\text{Minimal} = \phi$$

$$\text{Greatest} = \{a, b, c\}$$

$$\text{Least} = \phi$$

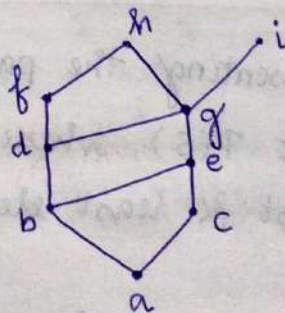


GLB, LUB:

GLB: Greatest Lower Bound {infimum} - Product / Meet - $\ast, \wedge, \cdot$

LUB: Least Upper Bound {supremum} - Sum / Join - $+, \vee, \oplus$

eg:



$$\text{UB of } \{a, b\} = \{b, d, e, f, g, h, i\}$$

$$\text{LUB of } \{a, b\} = \{b\}$$

$$\text{UB of } \{a, e, d\} = \{g, h, i\}$$

$$\text{LUB of } \{a, e, d\} = \{g\}$$

$$\text{LB of } \{d, g\} = \{d, b, a\}$$

$$\text{GLB of } \{d, g\} = \{d\}$$

$$LB \text{ of } \{c, d, g\} = \{a\}$$

$$GLB \text{ of } \{c, d, g\} = \{a\}$$

$$UB \text{ of } \{c, d, g\} = \{g, h, i\}$$

$$LUB \text{ of } \{c, d, g\} = \{g\}$$

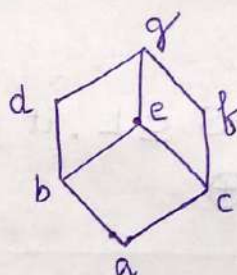
$$LB \text{ of } \{h, i\} = \{g, e, d, c, b, a\}$$

$$GLB \text{ of } \{h, i\} = \{g\}$$

28-02-25

Lattice : A lattice is a POSET $(L, \leq)$ which has every two elements $a, b \in L$ has GLB & LUB .

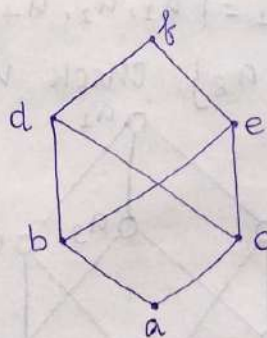
eg:



→ It a lattice

eg. for not a lattice :

Reason :



$$LB \text{ of } \{d, e\} = \{b, c, a\}$$

$$GLB \text{ of } \{d, e\} = \underline{DNE}$$

$$UB \text{ of } \{b, c\} = \{d, e, f\}$$

$$LUB \text{ of } \{b, c\} = \underline{DNE}$$

→ even if it does not satisfies for any of the elements, it is not a lattice.

Problems :

1. Is $(D_{30}, |)$ a lattice?

$$\text{Soln.: } D_{30} = \{1, 2, 3, 5, 6, 10, 15, 30\}$$

It is noted that for every

$$a, b \in D_{30}, a * b \in D_{30}$$

$$a + b \in D_{30}$$

Hence, it is a lattice.

Properties of lattice :

1. $a * a = a$ $a \oplus a = a$ (Idempotent)
2. $a * b = b * a$ $a \oplus b = b \oplus a$ (Commutative)
3. $(a * b) * c = a * (b * c)$ $(a \oplus b) \oplus c = a \oplus (b \oplus c)$ (Associative)
4. $a * (a \oplus b) = a$ $a \oplus (a * b) = a$ (Absorption)

Definition :

A lattice is an algebraic system $\langle L, *, \oplus \rangle$ with two binary operation $*$ and $\oplus$ on L which are both (1) commutative and (2) associative and satisfy the (3) absorption laws.

01-03-25

Sub-Lattice :

Let $\langle L, \oplus, * \rangle$ be a lattice. Let $S \subseteq L$, it is known as sub-lattice if it is closed on $*$ and $\oplus$.

Problems :

1. Let $\langle L, \leq \rangle$ be a lattices in which $L = \{a_1, a_2, \dots, a_8\}$ and S_1 , S_2 and S_3 be the subsets of L given by $S_1 = \{a_1, a_2, a_4, a_6\}$, $S_2 = \{a_3, a_5, a_7, a_8\}$ and $S_3 = \{a_1, a_2, a_4, a_8\}$. Check whether S_1 , S_2 & S_3 are sub-lattices or not.

Soln. :

In S_1 ,

It has been noted that

$$\forall a, b \in S_1 \Rightarrow a * b \in S_1$$

$$\Rightarrow a + b \in S_1$$

Hence, S_1 is a sub-lattice of L .

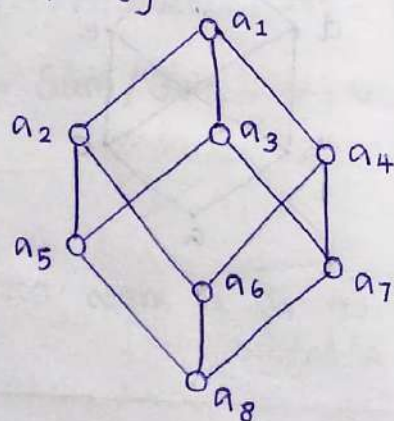
In S_2 ,

It has been noted that

$$\forall a, b \in S_2 \Rightarrow a * b \in S$$

$$\Rightarrow a + b \in S$$

Hence, S_2 is a sub-lattice.



In S_3 ,

$$a_2, a_4 \in S_3$$

$$a_2 * a_4 = a_6 \notin S_3$$

Hence, S_3 is not a sub-lattice of L .

Complete Lattice :

A complete lattice is such that every non-empty subset has GLB & LUB.

Bound Lattice :

A lattice which has both bound values (1, 0) - (greatest, lowest element), then the lattice is called bound lattice.

4-3-25

Complement Element :

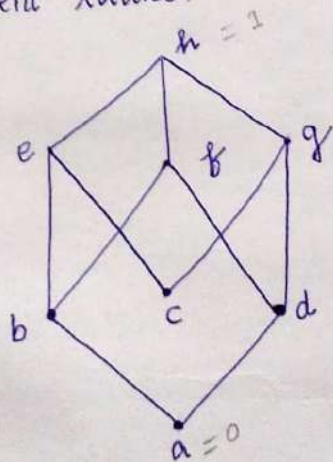
$$a * a' = 0$$

$$a + a' = 1$$

Here, a' is the complement of a .

Complemented Lattice :

If the complement of each element exists, then it is known as complement lattice.



Here,

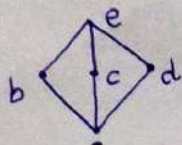
$$\left. \begin{array}{l} a' = f \\ b' = e \\ c' = f \\ d' = e \end{array} \right\} \begin{array}{l} \text{all elts. have} \\ \text{complement elts.} \\ \therefore \text{It is a} \\ \text{complement} \\ \text{lattice.} \end{array}$$

Distributive Lattice : $\rightarrow$ check with same level

$$\left. \begin{array}{l} \text{i) } a * (b + c) = (a * b) + (a * c) \\ \text{ii) } a + (b * c) = (a + b) * (a + c) \end{array} \right\} \text{for any 3 elts.}$$

Problems :

1. Show that the following lattice is not a distributive lattice.



Soln.: $b, c, d \in L$

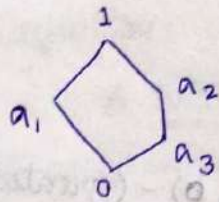
$$i) b * (c + d) = (b * c) + (b * d)$$

$$b * e = a + a$$

$$b \neq a$$

Hence, it is not a distributive lattice.

2. S.T the following lattice is not a distributive.



Soln.: $a_1, a_2, a_3 \in L$

$$a_1 * (a_2 + a_3) = (a_1 * a_2) + (a_1 * a_3)$$

$$a_1 * a_2 = 0 + 0$$

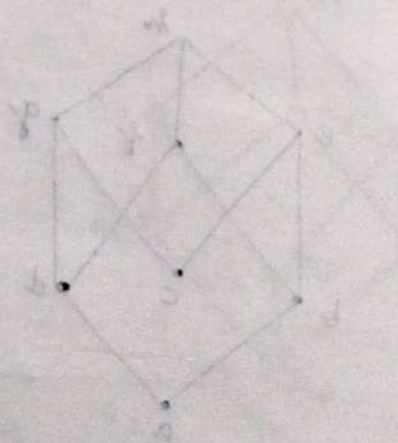
$$0 = 0$$

$$a_1, a_2, a_3 \in L$$

$$a_3 * (a_1 + a_2) = (a_3 * a_1) + (a_3 * a_2)$$

$$a_3 * 1 = 0 + a_3$$

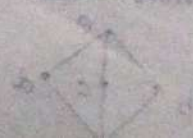
$$a_3 = a_3$$



complement of a is b
complement of b is a
It is a distributive lattice

Distributive Lattice:
i) $a * (b + c) = (a * b) + (a * c)$
ii) $a + (b * c) = (a + b) * (a + c)$

Problems:
1. Show that the following lattice is not a distributive lattice.



Boolean Algebra:

A lattice which is complemented & distributive it is a boolean algebra.

Sub Boolean Algebra:

Let B be a boolean algebra and S be $S \subseteq B$. If S contain elements 0 and 1 and is closed under $\langle +, * \rangle$ then S is known as sub boolean algebra of B .

Properties of Bounds:

$$a + a' = 1$$

$$a \cdot a' = 0$$

$$a + 0 = a$$

$$a \cdot 1 = a$$

$$a + 1 = 1$$

$$a \cdot 0 = 0$$

De-Morgan's Laws:

$$(a \cdot b)' = a' + b'$$

$$(a + b)' = a' \cdot b'$$

Note:

$$a + a'b = a + b$$

$$\begin{aligned} a + a'b &= (a + a') \cdot (a + b) \\ &= 1 \cdot (a + b) \\ &= a + b \end{aligned}$$

Problems:

1. Simplify the following boolean function. (Transform into SOP or POS). Also find its M^m notation.

$$i) ((x_1 x_2)' x_3)' ((x_1' + x_3)(x_2' + x_3'))'$$

Soln.:

$$\Rightarrow [(x_1 x_2)' x_3]' [(x_1' + x_3)(x_2' + x_3')]'$$

$$\Rightarrow [(x_1 x_2) + x_3'] [(x_1' + x_3)' + (x_2' + x_3')']$$

$$\Rightarrow [x_1 x_2 + x_3'] [x_1 * x_3' + x_2 x_3]$$

$$\Rightarrow x_1 x_2 x_1 x_3' + x_1 x_2 x_2 x_3 + x_3' x_1 x_3' + x_3' x_2 x_3$$

$$\Rightarrow x_1 x_2 x_3' + x_1 x_2 x_3 + x_1 x_3' + x_2 \cdot 0$$

$$\Rightarrow x_1 x_2 x_3' + x_1 x_2 x_3 + x_1 x_3' \cdot (x_2 + x_2') + 0$$

$$\Rightarrow x_1 x_2 x_3' + x_1 x_2 x_3 + x_1 x_2 x_3' + x_1 x_2' x_3'$$

$$\Rightarrow x_1 x_2 x_3' + x_1 x_2 x_3 + x_1 x_2' x_3'$$

$$= \sum m(6, 7, 4) = \sum m(4, 6, 7)$$

$$\text{ii) } x_1 (x_2' x_3)'$$

$$= x_1 (x_2 + x_3') = x_1 x_2 + x_1 x_3'$$

$$= x_1 x_2 (x_3 + x_3') + x_1 x_3' (x_2 + x_2')$$

$$= x_1 x_2 x_3 + x_1 x_2 x_3' + x_1 x_2 x_3' + x_1 x_2' x_3'$$

$$= x_1 x_2 x_3 + x_1 x_2 x_3' + x_1 x_2' x_3'$$

$$= \sum (7, 6, 4)$$

$$= \sum (4, 6, 7)$$

2. Simplify the following boolean expression:

$$\text{i) } x(y + z(xy + xz)')$$

$$\text{Soln.: } x(y + z(xy + xz)') = x(y + z(x'y')(x' + z'))$$

$$= xy + xzy(x'y')(x' + z')$$

$$= xy + x x' z y + x y' z y + x x' z y + x x' z y +$$

$$= xy + xz(x' + x'y' + x'z' + y'z')$$

$$= xy + 0 + 0 + 0 + 0$$

$$= xy$$

$$\text{ii) } xy' + z + (x' + y)z'$$

$$\text{Soln.: } xy' + z + (x' + y)z' = xy' + z + x'z' + yz'$$

$$= xy' + (z + z')(z + x') + yz'$$

$$= xy' + z + x' + yz'$$

$$= (x + x')(y' + x') + (z + y)(z + z')$$

$$= y' + x' + z + y$$

$$= 1 + x' + z = 1 + z$$

$$= 1$$

iii) Write in the form of SOP & its m-notation.

Canonical form

$$x(y' + z)$$

$$= xy' + xz$$

$$= xy'(z + z') + xz'(y + y')$$

$$= xy'z + xy'z' + xyz' + xy'z'$$

$$= xy'z + xy'z' + xyz'$$

$$= \sum m(5, 4, 6) = \sum m(4, 5, 6)$$

iv) Write in the form of POS & its M notation.
 Soln. : canonical form

$$x(y' + z')$$

$$= (x + yy')(y' + z' + xx')$$

$$= (x + y)(x + y')(x + y' + z')(x' + y' + z')$$

$$= (x + yz)(x + y + z')(x + y' + z)(x + y' + z')(x + y' + z)(x' + y' + z')$$

$$= (x + y + z)(x + y + z')(x + y' + z)(x + y' + z')(x' + y' + z)$$

$$= \Pi M(0, 1, 2, 3, 7)$$

$$v) a'b'c + ab'c + ab'c'$$

Soln. :

$$= b'c(a' + a) + ab'c'$$

$$= b'c(1) + ab'c'$$

$$= b'c + ab'c' = b'(c + ac')$$

$$= b'(c + a)(c + c')$$

$$= b'(c + a)$$

$$= b'c + ab'$$

$$vi) a'b(a' + c) + ab'(b' + c)$$

Soln. : $= a'b + a'b'c + ab' + ab'c$

$$= a'b(1 + c) + ab'(1 + c)$$

$$= a'b(1) + ab'(1)$$

$$= a'b + ab'$$

vii) $a + a'bc' + (b+c)'$

Soln. :

$$= a + a'bc' + (b+c)'$$

$$= a + a'bc' + b'.c'$$

$$= a + c'(a'b + b')$$

$$= a + [(a' + b').(b' + b)c'] \quad \because b' + b = 1$$

$$= a + [(a' + b').c']$$

$$= a + a'c' + b'c'$$

$$= (a + a').(a + c') + b'c'$$

$$= 1.a + (c' + b'c') = a + c'(1 + b')$$

$$= a + c'$$

$$\because 1 + b' = 1$$